Probabilistic Aspects of Diffusion-based Generative Models

Sotirios Sabanis (UoE/Archimedes/NTUA)

The University of Edinburgh & **Archimedes/Athena RC** & **National Technical University of Athens**

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Diffusion-based generative modes

- Generative modeling: Given a dataset of samples from an unknown distribution π_D, how to obtain samples distributed approximately from π_D?
- Diffusion-based/Score-based generative modeling:
  - Sequentially corrupting training data with slowly increasing noise

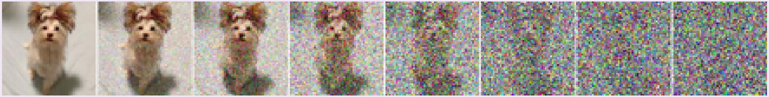

Figure: Noising process from [Song et al., 2021]

  - and then learning to reverse this corruption to form a generative model of the data
  - [Sohl-Dickstein et al., 2015], [Song and Ermon, 2019], [Ho et al., 2020]
  - State-of-the-arts results, e.g.
    - Image generation: [Dhariwal and Nichol, 2021], [Karras et al., 2022]
    - Audio and music generation: [Kong et al., 2021], [Mittal et al., 2021]
    - Realistic video generation: [Ho et al., 2022], [Harvey et al., 2022]
    - Molecular structure modeling: [Xu et al., 2022], [Trippe et al., 2023], [Watson et al., 2023].

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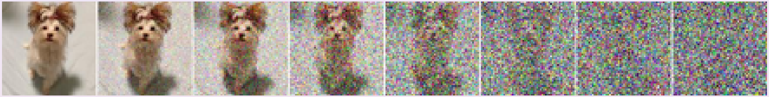

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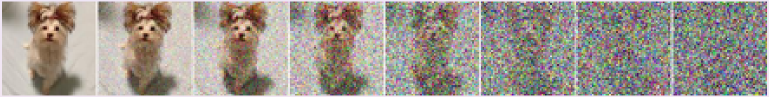

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Theoretical background

- In [Song et al., 2021], forward process $(X_t)_{t\in[0,T]}$ is an Ornstein-Uhlenbeck process

$$dX_t = -X_t\,dt + \sqrt{2}\,dB_t, \quad X_0 \sim \pi_D \in \mathcal{P}(\mathbb{R}^M), \quad (M \geq 1),$$

  - $(B_t)_{t\in[0,T]}$ is a Brownian motion (BM).

- [Anderson, 1982], [Haussmann and Pardoux, 1986], [Conforti et al., 2023]
- Backward process $(Y_t)_{t\in[0,T]} = (X_{T-t})_{t\in[0,T]}$:

$$dY_t = (Y_t + 2\nabla \log p_{T-t}(Y_t))\,dt + \sqrt{2}\,d\bar{B}_t, \quad Y_0 \sim \mathcal{L}(X_T),$$

  - $\{p_t\}_{t\in[0,T]}$ is the family of densities of $\mathcal{L}(X_t)$ w.r.t. the Lebesgue measure,
  - $\bar{B}_t$ is a BM independent of B_t.

- Implementation:
  1. $\widetilde{Y}_0 \sim \pi_\infty = \mathcal{N}(0, I_M)$ rather than $\mathcal{L}(X_T)$.
  2. Score function $\nabla \log p_t$ cannot be computed exactly.
  3. Numerical scheme to simulate backward dynamics (e.g. Euler–Maruyama scheme in [Song et al., 2021]).

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How to approximate $\nabla \log p_t$?

- $s : [0, T] \times \mathbb{R}^d \times \mathbb{R}^M \to \mathbb{R}^M$.
- Score-matching

$$\theta \mapsto \mathbb{E}\left[\int_0^T |\nabla \log p_t(X_t) - s(t, \theta, X_t)|^2 \, \mathrm{d}t\right].$$

- In practice,

$$\theta \mapsto U(\theta) := \int_\epsilon^T \frac{\kappa(t)}{T-\epsilon} \int_{\mathbb{R}^M} |\nabla \log p_t(x) - s(t, \theta, x)|^2 \, p_t(x) \, \mathrm{d}x \, \mathrm{d}t,$$

- $\epsilon > 0$ and $\kappa : [0, T] \to \mathbb{R}_{>0}$ is a weighting function.

- $\hat{\theta}$ is the (random) estimator of a minimiser θ^* of $U(\theta)$ obtained through some approximation procedure.

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$$= \mathbb{E}[\kappa(\tau)|\sigma_\tau^{-1} Z + s(\tau,\theta,m_\tau X_0 + \sigma_\tau Z)|^2] + C,$$

- $\tau \sim \mathrm{Unif}([\epsilon, T])$,
- $X_t \overset{\mathrm{d}}{=} m_t X_0 + \sigma_t Z$, with $Z \sim \mathcal{N}(0, I_M)$, $m_t = e^{-t}$, $\sigma_t^2 = 1 - e^{-2t}$.
- C independent of θ.

- Expression for $H : \mathbb{R}^d \times \mathbb{R}^m \to \mathbb{R}^d$:

$$H(\theta, \mathbf{x}) = 2\kappa(t) \sum_{i=1}^M \left(\sigma_t^{-1} z^{(i)} + s^{(i)}(t,\theta,m_t x_0 + \sigma_t z) \right) \nabla_\theta s^{(i)}(t,\theta,m_t x_0 + \sigma_t z),$$

- $\mathbf{x} = (t, x_0, z) \in \mathbb{R}^m$ with $m = 2M + 1$.

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Langevin samplers and discontinuous gradients

Task: Sample from a target distribution which is defined by

$$\pi_\beta(A) := \int_A \mathrm{e}^{-\beta u(\theta)}\,\mathrm{d}\theta \Big/ \int_{\mathbb{R}^d} \mathrm{e}^{-\beta u(\theta)}\,\mathrm{d}\theta,\ A \in \mathcal{B}(\mathbb{R}^d),$$

where $\beta \in \mathbb{R}_+$ is the 'inverse temperature parameter', $\mathcal{B}(\mathbb{R}^d)$ denotes the Borel sets of $\mathbb{R}^d$ and the function $u : \mathbb{R}^d \to \mathbb{R}_+$ is a continuous function with discontinuous gradient '$h := \nabla u$' .

Consider the so-called (overdamped) **Langevin** stochastic differential equation (SDE)

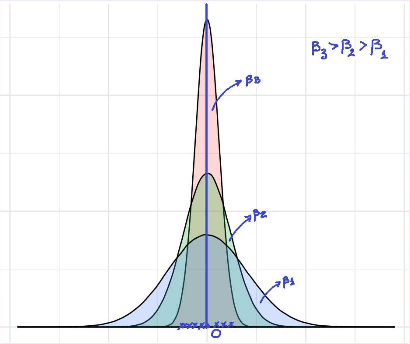
$$dL_t = -h(L_t)dt + \sqrt{2\beta^{-1}}dB_t, \tag{1}$$

where B is the standard Brownian motion in $\mathbb{R}^d$. Under suitable assumptions on h, e.g. semi-convexity & 'convexity at infinity' for u, the target measure (Gibbs measure) $\pi_\beta(\mathrm{d}\theta) \propto \exp(-\beta u(\theta))\mathrm{d}\theta$ is the unique invariant distribution of (1).

In particular, if one considers the following (linear) Langevin SDE (which is known also as OU process)

$$dL_t = -L_t dt + \sqrt{2\beta^{-1}} dB_t, \tag{2}$$

one knows that its limiting distribution is $\mathcal{N}(0, 1/\beta)$ with density function proportional to $e^{-\beta \frac{\theta^2}{2}}$. In other words, $u(\theta) = \frac{\theta^2}{2}$ and the (gradient) $h(\theta) = \theta$.



This is also true in the non-convex $u(\theta)$ case, i.e. π_β concentrates **around the minimizers** of u for sufficiently large β, see

C.-R. Hwang.
Laplace's method revisited: weak convergence of probability measures. The Annals of Probability.
8(6):1177–1182, 1980.

TheoPoula addresses both the well-known, well-documented exploding and vanishing gradient issues.

TAMED HYBRID ε-ORDER POLYGONAL UNADJUSTED LANGEVIN ALGORITHM

or, as an acronym, **THεO POULA**,

and has superior empirical performance (in given tasks) than widely adopted (adaptive) optimizers such as **Adam** (with more than 245k citations!) and some of its most popular variants, e.g. **AdamW**.

Diederik P. Kingma and Jimmy Ba.
Adam: A Method for Stochastic Optimization.
arXiv:1412.6980 [ICLR (Poster) 2015]

THEOPOULA (Daughter of God) vs ADAM (First Human)

Lim, D.-Y., and Sabanis, S.: Polygonal Unadjusted Langevin Algorithms: Creating stable and efficient adaptive algorithms for neural networks. Journal of Machine Learning Research 25 (53), 1–52.

Diffusion-based generative modelling in practice!

THEOPOULA (Daughter of God) **vs ADAM** (First Human Deucalion / Prometheus (Father / Titan))

Nanochat Pretraining

We pretrain the nanochat language model of

Karpathy, A.:
nanochat: The best ChatGPT that \$ 100 can buy,
GitHub repository, (2025).
https://github.com/karpathy/nanochat

at two depths, driving the matrix-valued parameters with the **boosted coordinate-wise** map of **Theopula** evaluated at the subgradient selection h. We refer to the **resulting scheme** as **SG-TheoPouLA**, and it attains the lowest validation bits per byte of the three optimisers at $\mathrm{L} = 12$, where every configuration was tuned directly due to the lack of available scaling laws, and remains competitive at $\mathrm{L} = 24$.

Table: Pretraining at $L = 12$, where lower is better for validation bits per byte and higher is better for CORE, with the best of each column in bold. AdamW drives the non-matrix groups in every row.

Optimiser	Tuning	Val bpb ↓	CORE ↑
Muon	Karpathy	0.8253	0.1664
Muon	ours	0.8199	0.1706
AdamW	Karpathy	0.8579	0.1143
AdamW	ours	0.8254	0.1523
SG-TheoPouLA (i)	ours	**0.8163**	0.1638
SG-TheoPouLA (ii)	ours	0.8199	**0.1715**

Table: Pretraining at $L = 24$, where the first two rows are published reference figures rather than runs of ours and the best among our runs is in bold. AdamW drives the non-matrix groups in every row.

Model or optimiser	Val bpb ↓	CORE ↑
OpenAI GPT-2	—	0.2565
Karpathy baseline	0.7186	0.2571
Muon	**0.7067**	**0.2637**
AdamW	0.7167	0.2577
SG-TheoPouLA	0.7133	0.2574

Lytras, I., Makras, N. and Sabanis, S.: The Tamed Subgradient Unadjusted Langevin Algorithm beyond Convexity. arXiv preprint arXiv:2608.06283 (2026).

Theoretical background - Denoising mechanism

- $\mathrm{d}Y_t = (Y_t + 2\nabla \log p_{T-t}(Y_t))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\bar{B}_t, \quad Y_0 \sim \mathcal{L}(X_T).$
- $\mathrm{d}\widetilde{Y}_t = (\widetilde{Y}_t + 2\,\nabla \log p_{T-t}(\widetilde{Y}_t))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\bar{B}_t, \quad \widetilde{Y}_0 \sim \pi_\infty = \mathcal{N}(0, I_M).$
- Novel process $(Y_t^{\mathrm{aux}})_{t\in[0,T]}$:

$$\mathrm{d}Y_t^{\mathrm{aux}} = (Y_t^{\mathrm{aux}} + 2\,s(T-t, \hat{\theta}, Y_t^{\mathrm{aux}}))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\bar{B}_t, \quad Y_0^{\mathrm{aux}} \sim \pi_\infty = \mathcal{N}(0, I_M).$$

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  - stepsize $\gamma \in (0,1)$, $\{\bar{Z}_k\}_{k\in\{0,\ldots,K+1\}}$ indep. M-dim standard Gaussian RVs.
- Cts-time interpolation $(\widehat{Y}_t^{\mathrm{EM}})_{t\in[0,T]}$ of Y_k^{EM}:

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- stepsize $\gamma \in (0,1)$, $\{\bar{Z}_k\}_{k\in\{0,\dots,K+1\}}$ indep. M-dim standard Gaussian RVs.

- Cts-time interpolation $(\widehat{Y}_t^{\mathrm{EM}})_{t\in[0,T]}$ of Y_k^{EM}:

$$\mathrm{d}\widehat{Y}_t^{\mathrm{EM}} = (\widehat{Y}_{\lfloor t/\gamma\rfloor\gamma}^{\mathrm{EM}} + 2\,s(T-\lfloor t/\gamma\rfloor\gamma, \hat{\theta}, \widehat{Y}_{\lfloor t/\gamma\rfloor\gamma}^{\mathrm{EM}}))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\bar{B}_t, \quad \widehat{Y}_0^{\mathrm{EM}} \sim \pi_\infty = \mathcal{N}(0, I_M).$$

- $\mathcal{L}(\widehat{Y}_k^{\mathrm{EM}}) = \mathcal{L}(Y_k^{\mathrm{EM}})$ at grid points for each $k \in \{0, \dots, K+1\}$.

Theoretical background - Denoising mechanism

- $\mathrm{d}Y_t = (Y_t + 2\nabla \log p_{T-t}(Y_t))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\bar{B}_t, \quad Y_0 \sim \mathcal{L}(X_T)$.
- $\mathrm{d}\widetilde{Y}_t = (\widetilde{Y}_t + 2\,\nabla \log p_{T-t}(\widetilde{Y}_t))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\bar{B}_t, \quad \widetilde{Y}_0 \sim \pi_\infty = \mathcal{N}(0, I_M)$.
- Novel process $(Y_t^{\mathrm{aux}})_{t\in[0,T]}$:

$$\mathrm{d}Y_t^{\mathrm{aux}} = (Y_t^{\mathrm{aux}} + 2\,s(T-t,\hat{\theta},Y_t^{\mathrm{aux}}))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\bar{B}_t, \quad Y_0^{\mathrm{aux}} \sim \pi_\infty = \mathcal{N}(0, I_M).$$

- Euler–Maruyama (EM) approximation $(Y_k^{\mathrm{EM}})_{k\in\{0,\ldots,K+1\}}$ of Y_t^{aux}:

$$Y_{k+1}^{\mathrm{EM}} = Y_k^{\mathrm{EM}} + \gamma(Y_k^{\mathrm{EM}} + 2\,s(T-k,\hat{\theta},Y_k^{\mathrm{EM}})) + \sqrt{2\gamma}\,\bar{Z}_{k+1}, \quad Y_0^{\mathrm{EM}} \sim \pi_\infty = \mathcal{N}(0, I_M).$$

- stepsize $\gamma \in (0,1)$, $\{\bar{Z}_k\}_{k\in\{0,\ldots,K+1\}}$ indep. M-dim standard Gaussian RVs.

- Cts-time interpolation $(\widehat{Y}_t^{\mathrm{EM}})_{t\in[0,T]}$ of Y_k^{EM}:

$$\mathrm{d}\widehat{Y}_t^{\mathrm{EM}} = (\widehat{Y}_{\lfloor t/\gamma\rfloor\gamma}^{\mathrm{EM}} + 2\,s(T-\lfloor t/\gamma\rfloor\gamma,\hat{\theta},\widehat{Y}_{\lfloor t/\gamma\rfloor\gamma}^{\mathrm{EM}}))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\bar{B}_t, \quad \widehat{Y}_0^{\mathrm{EM}} \sim \pi_\infty = \mathcal{N}(0, I_M).$$

- $\mathcal{L}(\widehat{Y}_k^{\mathrm{EM}}) = \mathcal{L}(Y_k^{\mathrm{EM}})$ at grid points for each $k \in \{0,\ldots,K+1\}$.

Theoretical background - Denoising mechanism

- $\mathrm{d}Y_t = (Y_t + 2\nabla \log p_{T-t}(Y_t))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\bar{B}_t, \quad Y_0 \sim \mathcal{L}(X_T).$
- $\mathrm{d}\widetilde{Y}_t = (\widetilde{Y}_t + 2\,\nabla \log p_{T-t}(\widetilde{Y}_t))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\bar{B}_t, \quad \widetilde{Y}_0 \sim \pi_\infty = \mathcal{N}(0, I_M).$
- Novel process $(Y_t^{\mathrm{aux}})_{t\in[0,T]}$:

$$\mathrm{d}Y_t^{\mathrm{aux}} = (Y_t^{\mathrm{aux}} + 2\, s(T-t, \hat{\theta}, Y_t^{\mathrm{aux}}))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\bar{B}_t, \quad Y_0^{\mathrm{aux}} \sim \pi_\infty = \mathcal{N}(0, I_M).$$

- Euler–Maruyama (EM) approximation $(Y_k^{\mathrm{EM}})_{k\in\{0,\dots,K+1\}}$ of Y_t^{aux}:

$$Y_{k+1}^{\mathrm{EM}} = Y_k^{\mathrm{EM}} + \gamma(Y_k^{\mathrm{EM}} + 2\, s(T-k, \hat{\theta}, Y_k^{\mathrm{EM}})) + \sqrt{2\gamma}\,\bar{Z}_{k+1}, \quad Y_0^{\mathrm{EM}} \sim \pi_\infty = \mathcal{N}(0, I_M).$$

- stepsize $\gamma \in (0,1)$, $\{\bar{Z}_k\}_{k\in\{0,\dots,K+1\}}$ indep. M-dim standard Gaussian RVs.

- Cts-time interpolation $(\widehat{Y}_t^{\mathrm{EM}})_{t\in[0,T]}$ of Y_k^{EM}:

$$\mathrm{d}\widehat{Y}_t^{\mathrm{EM}} = (\widehat{Y}_{\lfloor t/\gamma\rfloor\gamma}^{\mathrm{EM}} + 2\, s(T-\lfloor t/\gamma\rfloor\gamma, \hat{\theta}, \widehat{Y}_{\lfloor t/\gamma\rfloor\gamma}^{\mathrm{EM}}))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\bar{B}_t, \quad \widehat{Y}_0^{\mathrm{EM}} \sim \pi_\infty = \mathcal{N}(0, I_M).$$

- $\mathcal{L}(\widehat{Y}_k^{\mathrm{EM}}) = \mathcal{L}(Y_k^{\mathrm{EM}})$ at grid points for each $k \in \{0, \dots, K+1\}$.

Reference Semi-convex case

Bruno, S. and Sabanis, S.(in press). Wasserstein Convergence of Score-based Generative Models under Semiconvexity and Discontinuous Gradients. Transactions on Machine Learning Research (TMLR), arXiv:2505.03432.

Semi-convex case

The optimal value θ^* of the parameter θ is determined by optimizing the following score-matching objective

$$\mathbb{R}^d \ni \theta \mapsto \mathbb{E}\left[\int_0^T |\nabla \log p_t(X_t) - s(t, \theta, X_t)|^2 \, \mathrm{d}t\right]. \tag{3}$$

Assumption 1

Let θ^* be a minimiser[a] of (3) and let $\hat{\theta}$ be the (random) estimator of θ^* obtained through some approximation procedure such that $\mathbb{E}[|\hat{\theta}|^2] < \infty$. There exists $\widetilde{\varepsilon}_{AL} > 0$ such that

$$\mathbb{E}[|\hat{\theta} - \theta^*|^2] < \widetilde{\varepsilon}_{AL}.$$

[a] The score-matching optimization problem (3) is not necessarily (strongly) convex.

Subdifferentials

For any $x \in A \subseteq \mathbb{R}^d$ and any function $U : A \to \mathbb{R}$, the subdifferential $\partial U(x)$ of U at x is defined as

$$\partial U(x) = \left\{ p \in \mathbb{R}^d : \liminf_{y \to x} \frac{U(y) - U(x) - \langle p, y - x \rangle}{|y - x|} \geq 0 \right\}. \tag{4}$$

At the points where U is differentiable, it holds that $\partial U(x) = \{\nabla U(x)\}$.

For the sake of presentation, an element of $\partial U(x)$ is denoted by $h(x)$ for any $x \in \mathbb{R}^d$.

Assumption 2

The data distribution π_{D} has a finite second moment and it is absolutely continuous with respect to the Lebesgue measure with $\pi_{\mathrm{D}}(\mathrm{d}x) = \exp(-U(x))\,\mathrm{d}x$ for some $U : \mathbb{R}^d \to \mathbb{R}$. Moreover,

1. *The potential U is continuous and its gradient exists a.e.*
2. *The potential U is K-semiconvex. That is, there exists $K, R \geq 0$, such that for all $x, \bar{x} \in \mathbb{R}^d$,*

   $$\langle h(x) - h(\bar{x}), x - \bar{x}\rangle \geq -K|x - \bar{x}|^2, \qquad \text{when} \quad |x - \bar{x}| < R,$$

   or equivalently $U + \frac{K}{2}|\cdot|^2$ is convex.
3. *The potential U is μ-strongly convex at infinity. That is, there exists $\mu > 0$ and $R \geq 0$, such that for all $x, \bar{x} \in \mathbb{R}^d$,*

   $$\langle h(x) - h(\bar{x}), x - \bar{x}\rangle \geq \mu|x - \bar{x}|^2, \qquad \text{when} \quad |x - \bar{x}| \geq R. \tag{5}$$

Assumption 3

The function $s : [0, T] \times \mathbb{R}^M \times \mathbb{R}^d \to \mathbb{R}^d$ *is continuously differentiable in* $x \in \mathbb{R}^d$*. Let* $D_1 : \mathbb{R}^M \times \mathbb{R}^M \to \mathbb{R}_+$*,* $D_2 : [0, T] \times [0, T] \to \mathbb{R}_+$ *and* $D_3 : [0, T] \times [0, T] \to \mathbb{R}_+$ *be such that* $\int_\epsilon^T \int_\epsilon^T D_2(t, \bar{t})\, dt\, d\bar{t} < \infty$ *and* $\int_\epsilon^T \int_\epsilon^T D_3(t, \bar{t})\, dt\, d\bar{t} < \infty$*. For* $\alpha \in \left[\frac{1}{2}, 1\right]$ *and for all* $t, \bar{t} \in [0, T]$*,* $x, \bar{x} \in \mathbb{R}^d$*, and* $\theta, \bar{\theta} \in \mathbb{R}^M$*, we have that*

$$\begin{aligned} |s(t, \theta, x) - s(\bar{t}, \bar{\theta}, \bar{x})| \leq & D_1(\theta, \bar{\theta})|t - \bar{t}|^\alpha + D_2(t, \bar{t})|\theta - \bar{\theta}| \\ & + D_3(t, \bar{t})|x - \bar{x}|, \end{aligned}$$

where D_1*,* D_2 *and* D_3 *have the following growth in each variable: i.e. there exist* K_1*,* K_2*, and* $\mathrm{K}_3 > 0$ *such that for each* $t, \bar{t} \in [0, T]$ *and* $\theta, \bar{\theta} \in \mathbb{R}^M$*,*

$$|D_1(\theta, \bar{\theta})| \leq \mathrm{K}_1(1 + |\theta| + |\bar{\theta}|), \qquad |D_2(t, \bar{t})| \leq \mathrm{K}_2(1 + |t|^\alpha + |\bar{t}|^\alpha),$$
$$|D_3(t, \bar{t})| \leq \mathrm{K}_3(1 + |t|^\alpha + |\bar{t}|^\alpha).$$

Assumption 4

Let s be as in Assumption 3 and there exists $\mathrm{K}_4 > 0$ *such that, for all* $x, \bar{x} \in \mathbb{R}^d$ *and for any* $k = 1, \ldots d$,

$$|\nabla_x s^{(k)}(t, \theta, x) - \nabla_{\bar{x}} s^{(k)}(t, \theta, \bar{x})| \leq \mathrm{K}_4(1 + 2|t|^{\alpha})|x - \bar{x}|.$$

Assumption 5

There exists $\varepsilon_{SN} > 0$ *such that*

$$\mathbb{E} \int_0^{T-\epsilon} |\nabla \log p_{T-r}(Y_r^{aux}) - s(T - r, \hat{\theta}, Y_r^{aux})|^2 \, dr < \varepsilon_{SN}. \tag{6}$$

Theorem 1

Let Assumptions 1, 2, 3 and 5 hold. Then, there exist constants C_1, C_2, C_3 and $C_4 > 0$ such that for any $T > 0$ and $\gamma, \epsilon \in (0,1)$,

$$\begin{aligned} W_2(\mathcal{L}(Y_K^{EM}), \pi_{\mathrm{D}}) \leq & C_1\sqrt{\epsilon} + C_2 e^{-2\int_\epsilon^T \beta_t^{OS,K,\mu}\mathrm{d}t - \epsilon} \\ & + C_3(T,\epsilon)\sqrt{\varepsilon_{SN}} + C_4(T,\epsilon)\gamma^{1/2}, \end{aligned} \tag{7}$$

where C_1, C_2, C_3 and C_4 are given explicitly in the article.

Theorem 2

Let Assumptions 1, 2, 4 and 5 hold, and assume that $\mathbb{E}[|\hat{\theta}|^4] < \infty$ Then, there exist constants C_1, C_2, C_3 and $\widetilde{C}_4 > 0$ such that for any $T > 0$ and $\gamma, \epsilon \in (0, 1)$,

$$W_2(\mathcal{L}(Y_K^{EM}), \pi_{\mathrm{D}}) \leq C_1\sqrt{\epsilon} + C_2 e^{-2\int_\epsilon^T \beta_t^{OS,K,\mu}\,\mathrm{d}t-\epsilon} + C_3(T,\epsilon)\sqrt{\varepsilon_{SN}} + \widetilde{C}_4(T,\epsilon)\gamma^\alpha, \tag{8}$$

where C_1, C_2, C_3 and $\widetilde{C}_4$ are given explicitly in the article.

where is $\beta_t^{\mathrm{OS},K,\mu}$ estimated to be

$$\beta_t^{\mathrm{OS},K,\mu} = \frac{\mu}{\mu + (1-\mu)e^{-2t}} - \frac{e^{-2t}}{(\mu + (1-\mu)e^{-2t})^2}(K+\mu). \tag{9}$$

Moreover, it holds that

$$\lim_{t\to 0} \beta_t^{\mathrm{OS},K,\mu} = -K,$$

and

$$\lim_{t\to\infty} \beta_t^{\mathrm{OS}} = \lim_{t\to\infty} \beta_t^{\mathrm{OS},K,\mu} = 1,$$

which is consistent with $\pi_\infty \sim \mathcal{N}(0, I_d)$, the invariant distribution of the OU process.

Comment : Let U be K-semiconvex as in Assumption 2-(ii) and be μ-strongly convex at infinity as in Assumption 2-(iii) and fix $t \in (0, T]$. Then

$$\langle \nabla \log p_t(x) - \nabla \log p_t(\bar{x}), x - \bar{x} \rangle \leq -\beta_t^{\mathrm{OS}} |x - \bar{x}|^2, \qquad \text{for} \quad x, \bar{x} \in \mathbb{R}^d, \tag{10}$$

where

$$\beta_t^{\mathrm{OS}} = \frac{\mu}{\mu + (1-\mu)e^{-2t}} - \frac{e^{-2t}}{(\mu + (1-\mu)e^{-2t})^2} L, \tag{11}$$

for some $L > 0$ satisfying a 'weak convexity profile' (see [Conforti, 2024 and [Gentiloni-Silveri and Ocello, 2025]).

Numerical illustration

We provide a numerical illustration of the critical time when a non-log-concave distribution becomes log-concave in the case when π_{D} is a one-dimensional Gaussian mixture with two equi-weighted modes $\eta = 2$, each mode having same variance $s^2 = 9$, namely

$$\pi_{\mathrm{D}}(\mathrm{d}x) = (2(18\pi)^{1/2})^{-1} \left(\exp\left(-\frac{|x-2|^2}{18} \right) + \exp\left(-\frac{|x+2|^2}{18} \right) \right) \mathrm{d}x. \tag{12}$$

For this choice of π_{D}, the semiconvexity constant $K = \frac{2\eta^2}{(s^2)^2} = \frac{8}{81}$, and the strong convexity at infinity constant $\mu = \frac{s^2-2\eta^2}{(s^2)^2} = \frac{1}{81}$. Here, $R \geq 0$. The score function of 12 is given by

$$\nabla \log p_t(x) = -\frac{x}{9m_t^2+\sigma_t^2} + \frac{2m_t}{9m_t^2+\sigma_t^2}\frac{\exp\left\{\frac{-(x-2m_t)^2}{2(9m_t^2+\sigma_t^2)}\right\} - \exp\left\{\frac{-(x+2m_t)^2}{2(9m_t^2+\sigma_t^2)}\right\}}{\exp\left\{\frac{-(x-2m_t)^2}{2(9m_t^2+\sigma_t^2)}\right\} + \exp\left\{\frac{-(x+2m_t)^2}{2(9m_t^2+\sigma_t^2)}\right\}},$$
$$t \in [0, T],\ x \in \mathbb{R}, \tag{13}$$

where $m_t = e^{-t}$ and $\sigma_t^2 = 1 - e^{-2t}$ comes from the representation of the OU process. Figure 2 displays the time behaviour of score function 13 over the time interval $[0, 10]$ for fixed values $x = -0.8$ and $x = 0.5$. We also indicate the time $\bar{t} = \ln\left(\sqrt{1+\frac{K}{\mu^2}}\right) \approx 3.2377 < t^\star$ from Proposition 13. The score function in Figure 2 converges to $\nabla \log(\pi_\infty(x)) = -x$, where $\pi_\infty = \mathcal{N}(0, 1)$.

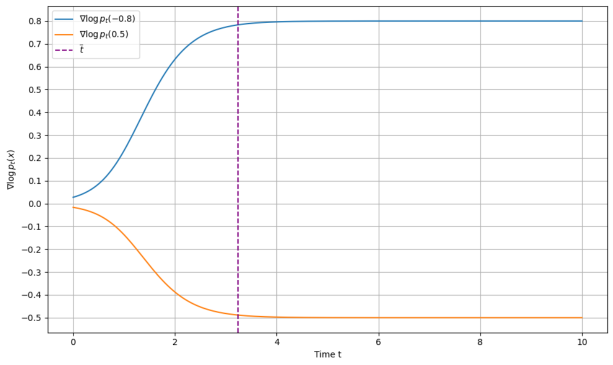

Figure: Score function 13 for fixed x and time $\bar{t}$.

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Thank you!